

Representing Content and Meaning in Discourse

The goal of the paper is to describe a new dynamic system in which we can explicitly represent both the static contents (i.e. propositions) and the dynamic meanings (i.e. context-change potentials) of sentences in discourse. I argue that natural language discourse and dialogue treats contents on a par with individuals – hence, we should represent and manage discourse reference to contents (propositions) just as we represent discourse reference to individuals.

The central observation is that modal subordination (e.g. *A wolf might come in. It would eat John*) provides a paradigm for retrieving static contents from dynamic meanings and determining relations between contents in a context-sensitive way. This new perspective makes possible an integrated analysis of several phenomena:

(a) entailment relations established *within discourse* by particles like *hence* or *therefore*, e.g. *A man saw a woman. Therefore, he noticed her.*;

(b) complex instances of modal subordination involving conjunction and both deontic and epistemic modalities, e.g. *You should buy a lottery ticket and put it in a safe place. (You're a person with good luck.) It might be worth millions.* (example from Roberts (1996));

(c) the parallel between *structured* plural anaphora in the individual and modal domains: consider the discourse *Every man saw a woman. Therefore, they noticed them.* – the entailment relation obtains only if the seeing and the noticing are structured in the same way, i.e. if the premise asserts that man m_1 saw woman n_1 and man m_2 saw woman n_2 , then we interpret the conclusion as asserting that m_1 noticed n_1 (and not n_2 !) and m_2 noticed n_2 (and not n_1 !); similarly, the entailment relation obtains in *A wolf might come in. It would see John. Therefore, it would notice him.* only if the pairing between (epistemic) possibilities and hypothetical wolves established in the premises is the same as the one in the conclusion, i.e. if we consider possibility p_1 in which wolf u_1 enters and sees John and possibility p_2 featuring wolf u_2 entering and seeing John, then the conclusion asserts that p_1 features the same wolf u_1 (and not u_2 !) noticing John – and similarly for p_2 and u_2 .

The system is formulated in many-sorted type logic (following the program in Muskens (1995, 1996)), which makes it possible to compare and integrate analyses from a variety of frameworks: (a) I incorporate in a dynamic framework the classical possible-worlds treatment of entailment as proposition inclusion and (b) I extend the treatment of individual-level plural anaphora in van den Berg (1996) and Nouwen (2003) with a parallel treatment of modal anaphora building on Stone (1999) and incorporating the possible-worlds account of modal quantification in Lewis (1973) and Kratzer (1981).

References:

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